

LLM Debugging & Monitoring

Both **Arize Phoenix** and **TruLens** are tools used in **LLMOps** to **evaluate, debug, and monitor LLM applications** like RAG pipelines, chatbots, and AI agents.

They help answer questions like:

-  Is my LLM giving correct answers?
-  Are retrieved documents relevant?
-  Is the model hallucinating?
-  Where did the pipeline fail (retriever, prompt, LLM)?

1 Arize Phoenix

What it is

Arize Phoenix is an **open-source LLM observability and evaluation platform** used to debug and analyze LLM applications.

Think of it as:

 **Grafana for LLM systems**

It shows **traces of the entire LLM pipeline**.

What Phoenix does

1 LLM Tracing

Shows every step of a request.

Example trace:

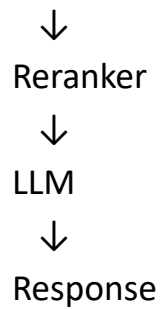
User Query



Retriever (Qdrant)



Embedding



Phoenix visualizes each step as **spans** so developers can see where problems occur.

2 LLM Evaluation

Phoenix can evaluate:

Metric	Purpose
Faithfulness	Detect hallucination
Correctness	Check factual accuracy
Document relevance	Evaluate retriever
Toxicity	Detect unsafe output

These evaluations use **LLM-as-a-judge models** to automatically score responses.

3 Dataset Analysis

Phoenix can cluster queries and responses using embeddings to detect poor performance patterns.

Example Use Case (RAG debugging)

If your pipeline is:

FastAPI



LangChain



Qdrant



Reranker



LLM

Phoenix helps answer:

- Did Qdrant retrieve the right docs?
 - Did reranker pick correct chunks?
 - Did the LLM hallucinate?
-

Simple Example

```
import phoenix as px
```

```
px.launch_app()
```

This launches a **local dashboard** showing traces of your LLM app.

2 TruLens

What it is

TruLens is a **Python library for evaluating LLM outputs using feedback functions**.

Instead of monitoring the pipeline, it **scores the final response quality**.

How TruLens Works

After an LLM response is generated:

User Query



LLM



Response



TruLens Feedback Functions



Evaluation Score

These feedback functions evaluate:

- Relevance
- Accuracy
- Coherence
- Safety

Example

```
from trulens_eval import Tru
```

```
tru = Tru()
```

```
feedback = relevance_feedback(query, response)
```

Result:

Relevance Score = 0.91

3 Phoenix vs TruLens

Feature	Phoenix	TruLens
Type	LLM observability platform	Evaluation library
Focus	Pipeline debugging	Response evaluation
Tracing	✓ Yes	✗ No

Feature	Phoenix	TruLens
RAG debugging	✓ Excellent	Limited
Metrics	Built-in evals	Custom feedback functions
UI Dashboard	✓ Yes	Minimal
Language	Python / TS	Python

🔗 When to Use Each

Use Phoenix if you want

- Debug full **RAG pipelines**
- Monitor **retriever performance**
- Track **LLM traces**
- Visual dashboards

Use TruLens if you want

- Evaluate **response quality**
- Score outputs automatically
- Run **evaluation experiments**

5 Real Production LLM Stack

Many companies combine both.

User



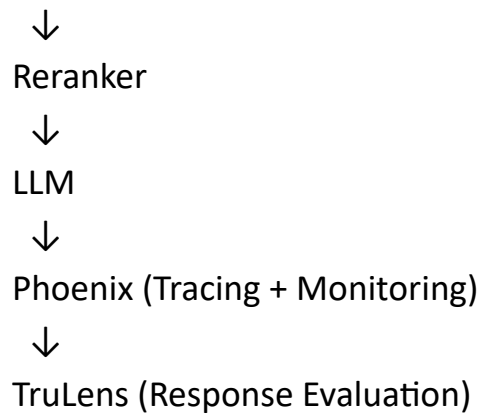
FastAPI



LangChain



Qdrant



✓ Simple rule

- Phoenix → Observability
- TruLens → Evaluation